Submission to the Joint Standing Committee on Electoral Matters Inquiry into the 2025 election

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List of recommendations

1. Compare the measured error rates against estimates of the margins. If the error rates are larger in any state or territory, investigate appropriately.
2. When sampling ballots from BPTCs for the assurance process, either select 5 ballots **at random** or select **all ballots** in the BPTC.
3. Use statistical inference methods that are designed for cluster samples rather than simple random samples. Most importantly, the calculation of the confidence intervals should be done for cluster samples.
4. Engage a qualified and independent statistical expert to: advise on the design of the whole assurance process; oversee correct implementation; and sign off on the final conclusions.
5. Commit to the (provisional) vote data publicly before the audit starts. Audit against the public data instead of the database (which cannot be scrutinised).
6. Use randomisation techniques that are transparent and reproducible.
7. Retain voting on paper ballots. Do not allow Internet voting or other forms of electronically assisted voting without a voter-verifiable paper record.

1 Introduction

The Australian election in 2025 was the second time in which an assurance process was conducted to assess the accuracy of the Senate vote digitisation process. Our submission relates to the conduct and results of this process. Henceforth, we also refer to this process as an 'audit', which is a concise and commonly used term for such procedures.

This audit substantially improves process and transparency, giving the Australian Electoral Commission (AEC) and all other Australians visibility into the accuracy of the counting process. However, the AEC and its appointed auditors still seem to have some gaps in their understanding of how the process should be conducted in order to have the best chance of detecting errors accurately and building public trust.

In 2022, the measured error rate in one state (Vic) was larger than the number of votes required to change who was elected (the "margin" of the election). In 2025, this happened in three states: Vic, NSW and WA—an illustration of the NSW case is shown in Figure 1. This does not mean that the outcome is wrong. It means that the error rate is large enough that the outcome *could*

be wrong. When this happens, it is important to investigate the nature of errors to ensure we can confidently determine that they won't change the election outcome. It is also worth working on improved digitisation processes with a smaller error rate.

Here is a brief summary of the topics we discuss in our submission:

- Section 2 – A comparison of margins and error rates for all states and territories in the 2025 Australian Senate election.
- Section 3 – Recommended improvements to the statistical design and analysis.
- Section 4 – Recommended improvements to the audit protocol.
- Section 5 – A reminder of reasons to avoid voting over the Internet.

Most of the material in our previous submission [BSS+22b] remains relevant. Further technical details are contained in [BSS+22a]. We have not repeated the details here, instead summarising this year's audit results and the implications.

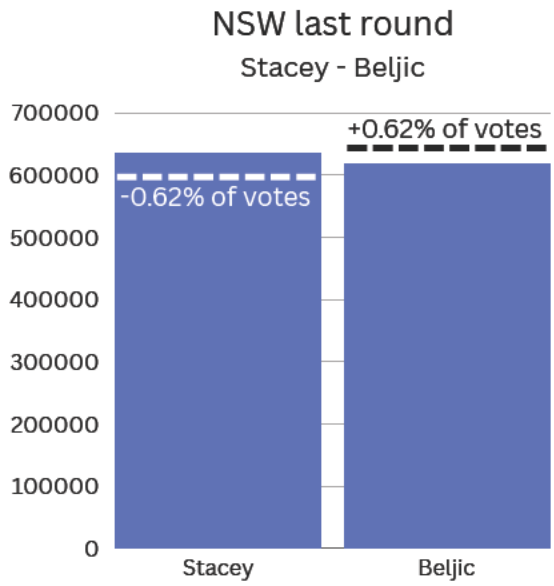

Figure 1: The last round of the 2025 NSW Senate election. The dashed lines show what the tallies would have been if the observed error rate (for all valid votes) was subtracted from the winner and added to the loser. Clearly this is sufficient to change who won.

2 Why we need to investigate error rates

When investigating possible errors in the counting of the votes, an important concept we need is the *margin* of an election. This is defined as the minimum number of votes that we need to change in order to arrive at a different winner or set of winners. In other words, it measures how "close" an election is.

For preferential voting systems, especially the one used for the Australian Senate, computing this kind of margin is difficult.

For the simpler preferential voting system used for the House of Representatives, an often used approximation for the margin is the *last-round margin*, which is the number of votes needed to change the outcome in the last round of the counting algorithm (the point at which one candidate first receives a majority, usually after some other candidates have been eliminated). This is easy to calculate: it is simply half the difference in votes (in the final round of counting) between the winning candidate

Table 1: Margin upper bounds, as calculated using ConcreteSTV, for each state and territory in the 2025 Australian Senate election. Each bound corresponds to possible change to the election outcome. Note that there may be smaller shifts that could also change the outcome but that were not detected by the algorithm. States in which the margin bound is smaller than the measured error rate are highlighted in yellow.

State/ Terr.	Formal votes	Margin bound	Margin bound (%)	Error rate (%)	Nature of the change to the outcome
ACT	293,474	23,194	7.90	0.08	Gallagher gets a quota.
NT	106,807	13,260	12.41	0.25	Price gets a quota.
TAS	371,790	5,746	1.55	0.48	Colbeck and Lambie get a quota.
VIC	4,101,762	16,021	0.39	0.51	Ananda-Rajah defeats Pickering; neither gets a quota.
NSW	4,986,832	8,664	0.17	0.62	Stacey defeats Beljic; neither gets a quota.
QLD	3,224,436	93,463	2.90	0.07	Roberts gets a quota.
SA	1,164,072	20,289	1.74	0.44	Walker gets a quota.
WA	1,622,016	4,185	0.26	0.52	Whitten defeats Singh; neither gets a quota.

and the one with the next highest tally. The last-round margin is often equal to the true margin, but otherwise provides an upper bound.

Unfortunately, for the Senate voting system the last-round margin is neither an upper bound nor a lower bound on the true margin. It may be that the actual margin is higher or lower than that value.

Sophisticated methods exist to compute possible minimal changes to the votes that would lead to different candidates being elected. Some of these are implemented in the ConcreteSTV software package. These methods give an upper bound on the true margin.

If the measured error rate in the digitisation process (i.e., as estimated from the audit), is greater than the upper bound on the election's margin (as a percentage of all ballots), then it is possible that the errors could have impacted who won.

Table 1 shows the measured error rates (from the audit), along with the smallest margin we were able to find, for each state and territory in the 2025 Australian senate election.[1] For three states, the error rates exceed the margin. As an example, Figure 1 illustrates the vote tallies of the two last-standing candidates in NSW and shows how errors could lead to a change of which one is elected.

Recommendation 1 *Compare the measured error rates against estimates of the margins. If the error rates are larger in any state or territory, investigate appropriately.*

It is not sound to assume there is no bias, or no systematic shift from some candidate(s) to others. If the AEC's analysis relies on such a claim, the discrepancies should be rigorously measured, and the results published to support the claim.

3 The Senate ballot paper assurance process lacks statistical rigour

This is the second Senate election for which an assurance process was conducted to examine the accuracy of the electronic data used for counting the votes, by comparing them to a sample of the physical ballot papers.

Since this process requires using statistical methodology, we are pleased to see that the AEC obtained advice from the Australian Bureau of Statistics (ABS) regarding the choice of sample sizes and what sampling rate would be required to achieve these.

[1] More details are available at: https://andrewconway.github.io/ConcreteSTV/FederalSenate2025/

Setting sample sizes is only one part of a statistical analysis. There are also choices to be made for other aspects of the sampling design (e.g., how the ballot papers should be sampled) and the subsequent analyses of the data (e.g., how to correctly calculate confidence intervals). Unfortunately, some of those steps were not carried out in a way that allows for valid or rigorous statistical conclusions. We describe below two specific examples of inappropriate analysis choices and frame recommendations that can address them.

These statistical shortcomings may stem from a lack of sufficient statistical expertise available to provide advice and oversight of this process. We expand on this below, and recommend that the AEC require that a qualified independent statistical expert be engaged in the design of the whole assurance process (not just the choice of sample sizes).

3.1 Errors in statistical methodology

At the point at which the ballot papers need to be sampled for the assurance process, they are stored as follows:

- Each ballot is stored in a *batch*, which can contain up to 50 ballots.
- Each batch is stored in a Ballot Paper Transport Container (BPTC), which can contain up to 10 batches.

To ensure valid statistical inference, the way the ballots are sampled needs to be randomised. Given the way the ballots are stored, it is practical to sample them in a "clustered" way, by first randomly selecting BPTCs and then randomly selecting ballots from each selected BPTC.

This is almost what was actually implemented, but the final step was not done. Rather, from each selected BPTC, only the first 5 ballots were selected, **without any randomisation**. Moreover, this detail was not specified in the methodology document published on the AEC website despite it being cruicial for statistical validity (we obtained the extra detail via direct correspondence with the AEC).

Unfortunately, the lack of randomisation substantially undermines what can be concluded from the assurance process. Since only about 10% of the ballots could possibly be sampled from (the first 5 ballots out of 50 across all BPTCs), any statistical conclusions would **only apply to those 10% of ballots**. Extending such conclusions to the remaining 90% of ballots requires assumptions that are not possible to test or verify from the data collected in the assurance process.

In other words, the estimates provided of the exception rates can only confidently be concluded for 10% of the ballots. The assurance process cannot confidently tell us whether those rates are similar when considering all ballots; it is possible that they are quite different.

Given that ballots are packaged into batches in a systematic way, rather than in a randomised order, it is plausible that scanning and digitisation error rates can also vary in a systematic way across ballots within a batch. Thus, assuming that the error rate is constant is not only untestable (with the current methodology) but also unsafe.

To fix this problem, we need to ensure that all ballots have a chance to enter into the sample. The following recommendation provides two possible choices.

Recommendation 2 *When sampling ballots from BPTCs for the assurance process, either select 5 ballots* ***at random*** *or select* ***all ballots*** *in the BPTC.*

There is a second fundamental error in the methodology that was used, and it is in the analysis of the data.

The way the confidence intervals were calculated was under the assumption that the ballots were sampled using what is known as a *simple random sample*. But this assumption is false. As discussed above, the ballots were collected using a cluster sample. Therefore, any analyses of the data needs to be done with more appropriate methods. As it stands, the confidence intervals provided are not reliable and may well be overconfident (too narrow). This same mistake was made in the assurance process at the previous Senate election in 2022.

Recommendation 3 *Use statistical inference methods that are designed for cluster samples rather than simple random samples. Most importantly, the calculation of the confidence intervals should be done for cluster samples.*

Like in the previous example above, the actual methods used for calculating the confidence interval were not specified in the published methodology document.

3.2 Need expert statistical oversight

The confidence that the assurance process is meant to provide rests on using appropriate statistical procedures. Therefore, it is vital that the someone with statistical expertise is involved in the design of the whole process, and also oversees its implementation to ensure it is followed and that any changes to the process in practice do not undermine statistical validity.

It appears that there is no such oversight at present. We say this based on the following evidence:

- The shortcomings we highlight above are basic and fundamental statistical errors.
- The methodology documents that were provided, and the report from the Service Provider who carried out the assurance process, lack key details about statistical methods.
- The advice that the AEC sought from the ABS is limited in scope and does not cover the whole process.

Engaging an expert could remedy all of these problems.

Recommendation 4 *Engage a qualified and independent statistical expert to: advise on the design of the whole assurance process; oversee correct implementation; and sign off on the final conclusions.*

4 Correcting the audit protocol

This section details process improvements to the audit protocol to ensure that the audit is a genuine test, from one end of the process to the other, of the accuracy of the Senate outcome.

The current process has two rather opaque steps that represent gaps in observability, allowing the possibility for a software or human error to go undetected.

The first is that the audit matches ballot papers against the database contents, rather than a fixed, public list of ballots. This means that a software error or security problem during the data export, or before the data export but after the audit, would not be detected by the audit. The AEC already publishes complete preference data to allow interested citizens to verify the count (which we have done). This step could simply be done earlier, in provisional form, as an initial setup for the audit. The audit can then be conducted as a comparison between the paper ballot and the published preferences, which can be counted by the public.

As a bonus, doing it this way would avoid the situation that occurred in 2022, in which the auditors were unable to find some votes in the database, but after investigation it was determined that these votes had been included (correctly) in the final published preferences. Simply auditing against the published preferences avoids either accidental omission or deliberate substitution.

This would not obstruct the iterative audit process—preferences could be published in batches as they are processed, with the audit occurring as the processing is finished, just as it does now.

Recommendation 5 *Commit to the (provisional) vote data publicly before the audit starts. Audit against the public data instead of the database (which cannot be scrutinised).*

A similar gap arises in the generation of the random sample. This year, the Service Provider simply used the `rand` function in an Excel spreadsheet. This does not give observers any evidence that the randomness was chosen fairly, and leaves open the risk that a poorly designed spreadsheet, misbehaving auditor, or malware on the auditor's device could have chosen a non-random sample. Simple techniques for transparent randomness generation are in our paper on Senate auditing [BSS+22a].

Recommendation 6 *Use randomisation techniques that are transparent and reproducible.*

5 Internet voting or paperless electronically assisted voting

Internet voting has been tried in other countries and some Australian states. Although some sophisticated techniques allow for some verification of some parts of the process, there is no usable solution that allows for a level of transparency that meets the security requirements of Australian elections. Any deployed system is likely to introduce a substantial risk of disruption or error.

Reliability is the most obvious problem. This issue caused thousands of voters to be disenfranchised and three Local Government Area elections to be voided in New South Wales in 2021 [CT22, Ful22, EC21]. Similar problems (concerning the automated delivery of Internet voting credentials by phone) led to the annulment of outcomes two seats of the French Assembly [CC23, CGGR23, DH23].

However, the most important problem is not obvious failures but *undetectable* error or manipulation caused by software errors or security vulnerabilities. Everything might seem to be fine, but actually be subject to systematic error that causes the wrong people to get elected.

We have written extensively on the security, privacy and verifiability of Internet voting systems [HT15, CRST15, HLPT20]. Scientists in other democracies have made similar findings. We omit most of the details here, but would be happy to discuss any of these issues with the committee.

Recommendation 7 *Retain voting on paper ballots. Do not allow Internet voting or other forms of electronically assisted voting without a voter-verifiable paper record.*

6 Conclusion

Elections must not only count the votes correctly, but provide transparent evidence to voters and scrutineers that the votes are counted correctly. The audit of Senate vote digitisation is a significant step forward, but needs improvement in order to fulfil the goal of providing rigorous evidence of an accurate count.

Internet voting, or electronically assisted voting without a voter-verifiable paper record, would put both the accuracy of the count and public trust in that accuracy at risk.

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